UNIT-I

WATER CHEMISTRY

Introduction

Water is the most useful and also the most abundant of all chemical substances. It plays an important role in wide variety of natural processes and is essential for the growth of all animals and plant life.

Water is basic material needed by almost all industries. Water is used as a solvent in chemical or pharmaceutical industries. Water is used in generating power in the form of steam and hydro-electric. A large quantity of water is also required in the industries like, paper, textile, sugar and chemical.

Common Impurities in Water:

The common impurities presents in water may be classified as follows.

1. Suspended Impurities.

Water may contain the suspended impurities such as clay, Mud algae, industrial waste, organic etc.

2. Dissolved Impurities.

i) Gases: The gases present in the air such as oxygen, carbon dioxide, nitrogen, oxides of Nitrogen, ammonia are soluble in natural water.

ii) Mineral salts: The commonly observed salts in natural water are carbonates, bicarbonates, sulphates and chlorides of calcium, magnesium, sodium and potassium.

3. Colloidal Impurities.

Colloidal particles of clay or other matter may be present in water. Such impurity neither settles down on standing is about 4-10 mm.

4. Biological Impurities.

The microbial impurities like algae, bacteria fungi are considered as pathogenic. Such impurities maybe present in natural water and may affect the efficiency of heat transfer equipments. Biological impurities can also contribute its usefulness.

water consumption by introducing in-factory water recycling systems in combination with wastewater treatment systems.

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1. Hardness

Harness is the sum of calcium and magnesium, which is a component of scale in combination with pH and methyl-red alkalinity. The scale is deposited on the heat exchange conductors in cooling equipment, and thus, causes impairment such as reduced heat conductivity.

Hardness of Water :

Definition of Hardness*:*

Hardness was originally defined as the soap consuming capacity of water sample. Also the characteristics of water which prevents lathering of soap is known as hardness of water.

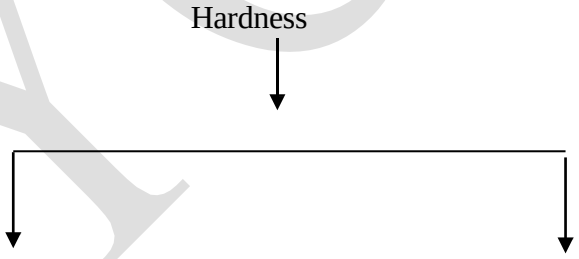
Soaps are generally considered as the sodium or potassium salts of long chain fatty acids such as oleic acid, palmetic acid or steric acid. The soap consuming capacity of water is mainly due to the presence of calcium, magnesium and other heavy metal ions. When sample of hard water (Containing these ions) reacts with the sodium salts of long chain fatty acids present in the soap to form insoluble white precipitate which do not have any detergent value.

$2\ C_{17}H_{35}COONa + Ca(HCO_3)_2 \longrightarrow (C_{17}H_{35}COO)_2Ca + 2\ NaHCO_3$

(Sodium Stearate) (Hard Water) Calcium Stearate (Scum)

Thus, hard water does not produce lather easily whereas lather is formed easily in soft water. Hence the hardness of water is considered as a measure of salts content in water.

Types of Hardness:



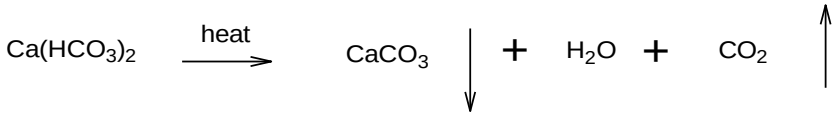
Temporary Hardness
(Bicarbonate and Carbonate of Ca^{++} and Mg^{++})

Permanent Hardness
(Cl^-, SO_4^{--}, NO_3^- of Ca^{++} & Mg^{++})

Depending upon the types of salts presents in water the hardness is broadly classified as.

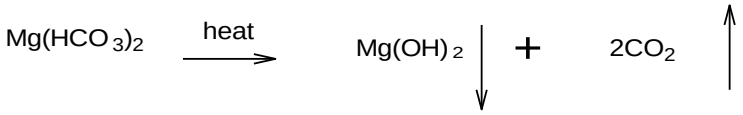
A) Temporary Hardness.

Temporary hardness is caused by the presence of dissolved carbonates & bicarbonates of calcium, magnesium and other heavy metals. It can be removed by boiling the water, when bicarbonates are decomposed, yielding insoluble carbonates or hydroxide, which are deposited as ppt at the bottom of vessel.



Calcium Bicarbonate (Sodium in water)

Calcium Carbonate (Insoluble in water)



Magnesium Bicarbonate (Soluble in water)

Magnesium hydroxide (Insoluble in water)

B) Permanent Hardness.

It is due to the presence of dissolved chlorides, sulphate and nitrates of calcium, magnesium, iron and other heavy metals. Permanent hardness is also known as non-carbonate or non- alkaline hardness. Unlike temporary, on boiling, permanent hardness cannot be removed. It requires special chemical methods.

C) Total Hardness = Temporary hardness + Permanent Hardness

Degree of Hardness: Calculation of $CaCO_3$ equivalent.

Hardness of water is conveniently expressed in terms of equivalent amount (equivalent) of $CaCO_3$. The reasons for choosing $CaCO_3$ as the standard for reporting hardness of water is the ease in calculations as its molecular weight is exactly 100 (equivalent weight =50) moreover it is the most insoluble salt that can be precipitated in water treatment .Therefore it is necessary to convert all the hardness producing salts in to their $CaCO_3$ equivalent .Hence $CaCO_3$ equivalent of substances is given by.

$$CaCO_3 \text{ equivalent} = \frac{\text{Weight of subtance}}{\text{Equivalent of weight of subtances}} \times 50$$

Or

$$CaCO_3 \text{ equivalent} = \frac{\text{weight Of Substance}}{\text{Molecular weight of substance}} \times 100$$

2. Units of Hardness:

I) Parts per million (ppm)

It is defined as the number of parts by weight of $CaCO_3$ equivalent per million (10^6) parts by weight of water.i.e.1ppm = 1 part of $CaCO_3$ equivalent hardness in 10^6 parts of water.

ii) Milligrams per liters (mg/lit)

It is defined as the number of milligrams of $CaCO_3$ equivalent hardness present in one litre

of water.

1mg/l = 1mg of $CaCO_3$ equivalent hardness per litre of water

iii) Clarke's degree (^{0}Cl)

It is defined as the number of parts by weight of $CaCO_3$equivalent hardness per 70,000 parts of water.

1^0Cl = 1 Part by weight of $CaCO_3$ equivalent hardness per 70,000 parts of water.

iv) Degree French (^{0}Fr)

It is defined as the number of parts by weight of $CaCO_3$equivalent hardness per 10^5 parts of water.

1^0 Fr = 1 Part of $CaCO_3$ equivalent hardness per 10^5 parts of water.

Relation between all units is as follows.

1 ppm = 1 mg/lit = 0.07^0C = 0.1^0Fr

3. Sterilization:

The process of destroying the disease–causing bacteria and micro-organisms etc. from water is known as disinfection or sterilization of water.

A) Boiling.

B) Chlorination.

1. By using chlorine, (Cl_2)
2. By using bleaching power, ($CaOCl_2$)
3. By using Chloramines, ($ClNH_2$)

C) Ozonisation (O_3).

D) Ultra-Violet Rays.

B) Chlorination:

Chlorine is the most common disinfecting agent in water treatment. It is capable of removing *E. coli* and substantially reducing other bacteria. Chlorine may be added in the form of bleaching powder or directly as a gas or in the form of concentrated solution in water. Whatever may be the method employed the treatment should give accurate dosage, good distribution and sufficient time of contact so as to ensure effective disinfection.

By using Chlorine.

Chlorine is most commonly used disinfectant in water treatment. It can be employed directly as a gas or in the form of concentrated solution in water. The chemical action of

chlorine is the production of hypochlorous acid which is a powerful germicide.

$$Cl_2 + H_2O \longrightarrow HOCl + HCl$$

$$HOCl \longrightarrow HCl + [O]$$

Nascent Oxygen

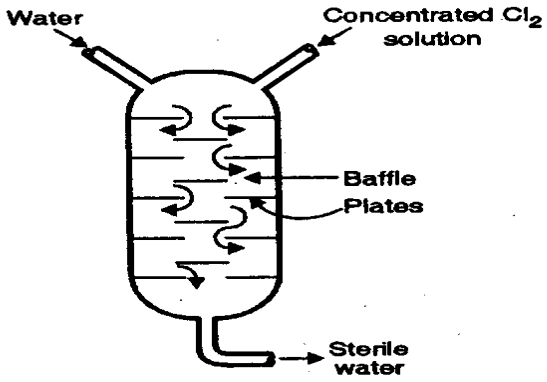
$$HOCl \rightarrow H^+ + OCl^-$$

Hypochlorite ions

It is earlier believed that the disinfecting action of chlorine was due to the nascent oxygen liberated, which oxidizes harmful bacteria etc. However after long research it was reported that the death of micro-organism, bacteria etc. results from chemical reaction of enzyme which is essential for the metabolic process of micro-organism, so the death of micro-organisms results due to inactivation of enzyme by **hypochlorous acid**.

The chlorine is found to be more effective disinfectant at lower pH value < 6.5

Liquid chlorine is most effective when applied to filtered water at such a point where adequate mixing is done. Apparatus used is called chlorinator which is a high tower having a no. of baffle plates. Water & proper quantity of concentrated chlorine solution are introduced at the top. During their passage, they get thoroughly mixed & the treated water is taken out from the bottom. For filtered water, 0.3 - 0.5 ppm of Cl_2 is sufficient.



Chlorinator

Factors affecting efficiency of Chlorine

1. Time of contact: Death rate is maximum in the starting & goes on decreasing with time.
2. Temperature Of water: The rate of reaction with enzyme increases with temperature.
3. P^H value of water: At lower pH values (5-6) , the rate of death is more.

Advantages of Chlorination

1) Effective & economical
2) Requires very little space for storage
3) Can be used at low as well as high temperature.
4) Stable & does not deteriorate on keeping.
5) Does not introduce any salt impurities in water.
6) It is the most ideal disinfectant.

Disadvantages of Chlorination

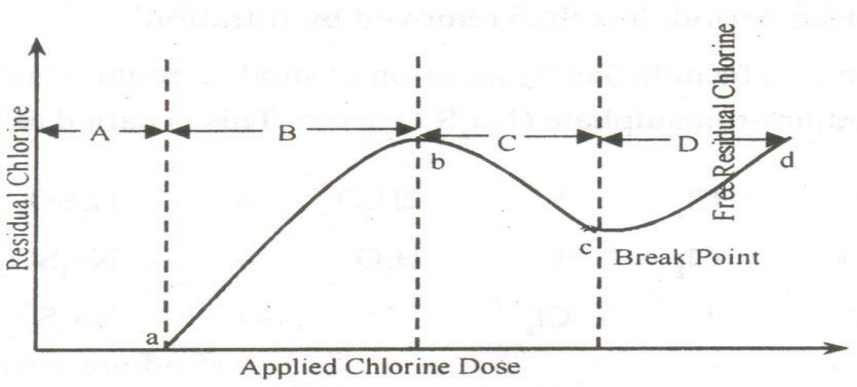
1) Excess of chlorine, if added, produce unpleasant taste & odour. Also produces irritation on mucus membrane. Therefore its dose should not be more than 0.1 – 0.2 ppm and pH of water should be below 6.5.

Break point Chlorination:

It involves the addition of sufficient amount of chlorine to oxidize

a) Organic matter
b) Reducing substances, and
c) Free ammonia

in raw water leaving behind mainly free chlorine which possess disinfecting action against disease producing bacteria (pathogens). When the doses of applied chlorine to water rich in organic compounds or ammonia is gradually increased, the results obtained can be shown graphically in which four stages appear.



Break Point Chlorination Curve

A. Chlorine consumption in oxidizing organic matter, reducing substances & ammonia

B. Formation of chloro-organic compounds and chloramines

C. Destruction of chloro-organic compounds and chloramines.

D. Free residual chlorine begins to appear.

The addition of chlorine at the dip or break is called break point chlorination.

This indicates the point at which free residual chlorine begins to appear.

When a small dose of chlorine is added to certain volume of water sample and shaken well after few minutes the chlorine gets consumed. If the chlorine dose is kept increasing in the same volume of water, after certain dose, it is found that certain amount of free chlorine is found present in water and it is called as residual chlorine. Out of the dose added some amount of chlorine is used for oxidation of reducing substances in water, killing of bacteria and chlorination of certain organic matter in the water.

Advantages of break point chlorination.

1. The break point chlorine dose oxidises reducing compounds, microorganism and chlorinates organic matter in water.
2. This dose removes colour from water and any dirty smell.
3. This dose prevents growth of bacteria, algae, fungi and weeds in water.
4. It destroys completely (100%) all the disease producing bacteria.

Softening of Water:

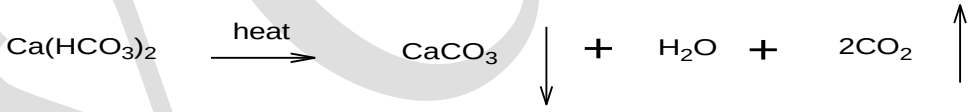
The process of removing hardness producing salts from water is known as softening of water.

Removal of Temporary Hardness:

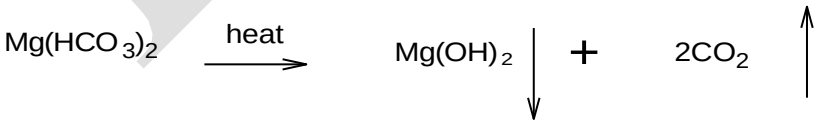
Temporary hardness is mainly due to bicarbonates of calcium and magnesium.

It can be removed by following method.

Boiling Method .



Calcium Bicarbonate (Sodium in water) — Calcium Carbonate (Insoluble in water)



Magnesium Bicarbonate — Magnesium hydroxide

i) Zeolite (Na_2Z Or Na_2Ze) or Permutit process of water softening

This is the modern process used for removing both temporary and permanent hardness of water. The chemical structure of zeolite or permutit is represented as below. $Na_2O.Al_2O_3$. x SiO_2. yH_2O Where x = 2-6 ,y =2-10

Hydrated Sodium–Alumino Silicates

It is represented as Na_2Z (where Z = Zeolite) Zeolite as sodium ions and can easily exchange their Na^+ ions with the hardness producing ions like Ca^{2+}, Mg^{2+} etc. from water.

Principle:

This is the modern process in which water is softened by passing it through sodium zeolite bed where Ca^{2+} and Mg^{2+} ions from the water exchange with equivalent amount of Na^+ ions of Zeolite.

Process:

Zeolite softener consists of a steel tank packed with a thick layer of zeolite. It is as shown in the following figure.

Zeolite softener

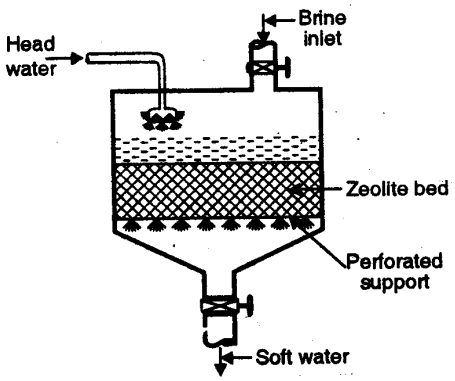
Hard water is passed through the zeolite, where the hardness producing ions (like Ca^{2+}, Mg^{2+} etc.) gets exchanged with Na^+ ions in the form of insoluble salts of calcium and magnesium zeolite. The resulting soft water is collected from the bottom of the tank.

The reactions taking place during the softening process as follows.

Reaction for Temporary hardness.

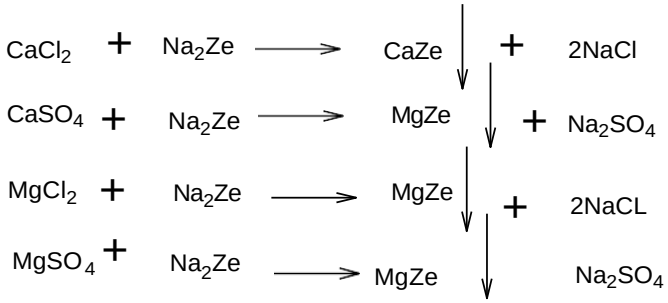
$$Ca(HCO_3)_2 + Na_2Ze \longrightarrow CaZe \downarrow + 2NaHCO_3$$

$$Mg(HCO_3)_2 + Na_2Ze \longrightarrow MgZe \downarrow + 2NaHCO_3$$



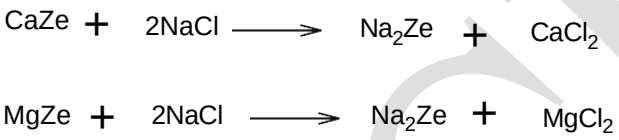
Zeolite Softener

Reaction for Permanent hardness.



Regeneration

After a long time (12hours) when the Zeolite is complete converted to Ca and Mg Zeolite it gets exhausted i.e. it stops the softening of water. However the zeolite can be regenerated by passing 10% brine (NaCl solution) from the exhausted zeolite for few minutes.



Exhausted Zeolite Regenerated Zeolite

The washing containing ($CaCl_2$ and $MgCl_2$) is drain out at the bottom and the regenerated zeolite bed (Na_2Ze), thus obtained is used again for softening of hard water.

Advantages of Zeolite process.

1. Hardness is removed up to 10ppm.
2. Equipment used is compact and occupies less space.
3. It automatically adjusts itself to water of different harness.
4. No sludge formation during the process.
5. Requires less time for softening.
6. It is quite clean.
7. It requires less skill for maintenance as well as operation.

Limitations of Zeolite process.

1. High turbidity water cannot be treated efficiently by this method, because fine impurities get deposited on the Zeolite bed ,thereby creating problem for its working.
2. If hard water contains mineral acid it may destroy zeolite.
3. Acid radicals like cl^-,$SO4^{--}$ are not removed in this process.
4. Hot water cannot be used because the Zeolite dissolves in hot water.
5. If hard water contain large quality of Mn^{2+} or Fe^{2+} ions then these ions form MnZe or FeZe respectively this cannot be regenerated easily.

Disadvantages of Zeolite process.

1. The treated water contain more sodium salts than in lime soda process.

2. When such softened water containing $NaHCO_3$, Na_2CO_3 etc) is used in boilers for steam generation, sodium bicarbonate decomposes producing CO_2 which causes corrosion and caustic embrittlement.

$$NaHCO_3 \longrightarrow NaOH+CO_2$$

Formula for calculating hardness of water using the Zeolite process:

$$\text{Hardness of water} = \frac{\text{Eq. wt. of } CaCO_3 \text{ x conc. of NaCl (gm/L) x vol. of NaCl (L) x 1000}}{\text{Eq. wt. of NaCl x vol. of water softened by Zeolite (L)}}$$

$$\text{Hardness of water} = \frac{\text{Eq. wt. of } CaCO_3 \text{ x \% of NaCl x vol. of NaCl (L) x 10000}}{\text{Eq. wt. of NaCl x vol. of water softened by Zeolite (L)}}$$

$$\text{Hardness of water} = \frac{\text{Eq. wt. of } CaCO_3 \text{ x conc. of NaCl (mg/L) x vol. of NaCl (L)}}{\text{Eq. wt. of NaCl x vol. of water softened by Zeolite (L)}}$$

ii) Ion Exchange or Deionization or Demineralization process:

In boiler pure water free from all types of impurities is required. Water of such quality can be obtained by demineralization process. It removes almost all the ions (Cation and anions) present in water. For this purpose ion- exchange resins are used. These are insoluble cross linked long chain organic polymers with micro porous structure. The functional group attached to the chains are responsible for the ion exchanging property

Principle

In ion-exchange process the water is softened by a reversible exchange of ions between stationary phase (ion exchange resin) and external mobile phase (water).

Cation Exchange resins (RH_2)

Resins containing acidic functional groups (-COOH, SO_3) are capable of exchanging their H^+ ions with other cations which come in their contact. These are represented as RH_2 so their exchanges reaction with cation (eg.Ca^{2+},Mg^{2+} can be represented as below

$$RH_2 + Ca^{++} \longrightarrow RCa + 2H^+$$

The impure hard water is passed first through cation exchanges which remove all the cations like Ca^{2+}, Mg^{2+}, Na^+ etc. from it an equivalent amount of H^+ ions are released.

$$RH_2 + Ca^{++} \longrightarrow RCa + 2H^+$$

$$RH_2 + Mg^{++} \longrightarrow RMg + 2H^+$$

$$RH_2 + 2Na^+ \longrightarrow RNa_2 + 2H^+$$

The H^+ ions then combine with the ions like chloride, sulphate and carbonate present in the water and gets converted in to the corresponding acids like HCl, H_2SO_4 and H_2CO_3. In other words, water from cation exchanger is free from all cations but is acidic in nature.

The acidic water is then passed through the anion exchange resins, where acidic ions are converted to water.

Anion Exchange resins. R'$(OH)_2$

Resins containing basic functional groups (-NH_2, =NH) are capable of exchanging their anions with other anions which come in their contact.

These are represented as R'$(OH)_2$ so they exchange OH^- ions with anions.

eg. SO_4^{--}, CO_3^{--},Cl^-)

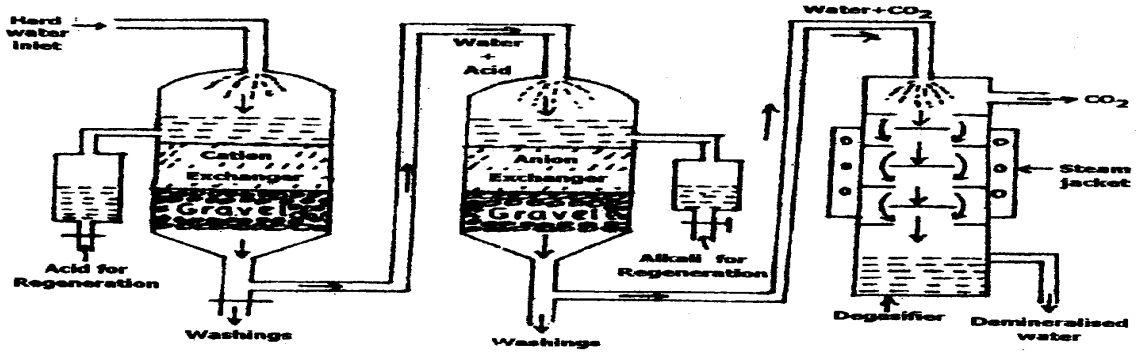
$$R'(OH_2) + SO_4^{--} \longrightarrow R'SO_4 + 2OH^-$$

$$R'(OH_2) + 2HCl \longrightarrow R'Cl + 2H^+$$

$$R'(OH_2) + H_2SO_4 \longrightarrow R'SO_4 + 2H^+$$

$$R'(OH_2) + H_3CO_3 \longrightarrow R'CO_3 + 2H^+$$

$$2H^+ + 2OH^- \longrightarrow 2H_2O$$



Regeneration -:

When the capacities of resins to exchange ion are lost, then they are said to be exhausted. The exhausted cation exchanger is regenerated by passing dilute HCl solution through cation exchanger.

The reactions in regeneration can be represented as below.

$$RCa + 2HCl \longrightarrow RH_2 + CaCl_2$$

$$RMg + H_2SO_4 \longrightarrow RH_2 + MgCl_2$$

Exhausted Cation Exchanger — Regenerated Cation Exchanger — (Washing)

The washing containing $CaCl_2$ and $MgCl_2$ is removed from the bottom of column.

Similarly the exhausted anion exchanger is regenerated by passing dilute solution. NaOH through anion exchanger. The reaction for regeneration can be represented as below.

$$R'Cl_2 + 2NaOH \longrightarrow R'(OH_2) + 2NaCl$$

$$R'SO_4 + 2NaOH \longrightarrow R'(OH_2) + Na_2SO_4$$

The washing containing NaCl and Na_2SO_4 is removed for the bottom of anion exchanger column.

Advantages of Ion–Exchange Process

1) This process can be used to soften highly acidic or alkaline water.
2) It produces water of very low hardness (about 2ppm) hence it is very good process of water softening.
3) Treated water contains negligible amount of total dissolved solids.
4) Water obtained can be used in high pressure boilers.
5) Method self adjusts with water of all types & different hardness.

Disadvantages of Ion –Exchange Process

1) The equipment is costly and more expensive chemical are needed.
2) If water contains turbidity, then the efficiency of process is reduced. Turbidity must be below 10 ppm.
3) Initial investment is high.
4) Method cannot be used for getting deionized water on large scale.

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Comparison between Lime soda process, Zeolite Process & Ion exchange process

Sr. No.	Zeolite Process	Ion exchange process
1	This is the modern process in which water is softened by passing it through sodium zeolite bed where Ca^{2+} and Mg^{2+} ions from the water exchange with equivalent amount of Na^{+} ions of Zeolite.	In ion-exchange process the water is softened by a reversible exchange of ions between stationary phase (ion exchange resin) and external mobile phase (water).
2	(Add reactions involved.)	(Add reactions involved.)
3	Capital cost is higher but operation expenses are lower.	Capital cost is higher & maintenance cost is higher
4	Water of generally, 8-10ppm residual hardness is obtained	Water of generally,0-2 ppm residual hardness is obtained
5	Treated water contains larger amounts of sodium salts	Treated water contains almost zero total dissolved salts
6	Self adjusts with different hardness of water.	Self adjusts with different hardness of water
7	Involves no precipitates	Involves no precipitates
8	Acidic water cannot be treated	Acidic water can be treated
9	High turbidity water affects the process of softening.	High turbidity water affect the process of softening.
10	It can operate under pressure & can be made fully automatic.	It can operate under pressure & can be made fully automatic.

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Boiler Troubles: -

The water which is used to generate steam in boilers is called as boiler feed water.

It should not contain matter in suspended or dissolved form. The other materials that are not desirable in boiler water are silica, oil and other dissolved gases.

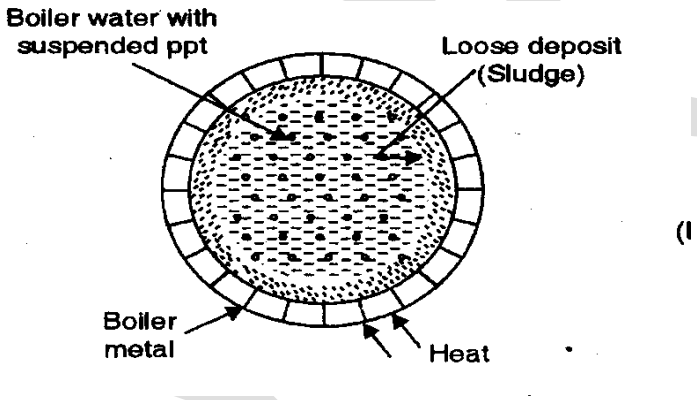
If the hard water is directly fed to the boiler then it raises many problems to the boiler called as **boiler trouble**. The main boiler troubles are.

1. Scale and Sludge formation.
2. Boiler corrosion.
3. Priming and Foaming.
4. Caustic Embrittlement.

- **Sludge and Scale formation:**

Sludge: *In boiler water is continuously converted in to steam. This results in increase in the concentration of dissolved salts. When the conc. reaches to saturation point, they get thrown out of water in the form of precipitates on the inner walls of the boiler.*

Sludge is loose, soft and muddy or slimy precipitate formed in the water of boiler. It is due to $MgCO_3$, $CaCO_3$ and $MgCl_2$etc.It can be represented as below.



Disadvantages / Drawbacks of Sludge formation.

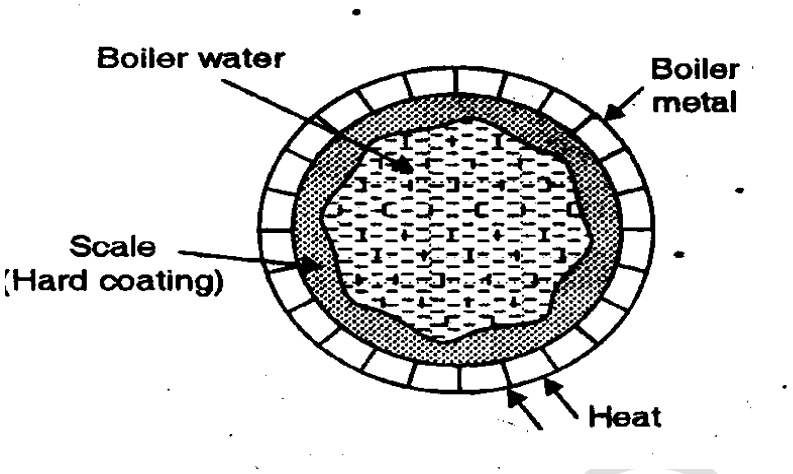
1. Sludges are poor conductors of heat, so they consume some portion of heat generated.
2. When Sludges are formed along with scales, then Sludges gets converted in to scales and both deposit as scales.
3. Water circulation gets disturbed, if pipes, valves and plug opening get chocked due to Sludges.

Prevention of Sludge formation.

1) By using well softened water.
2) By frequent blow –down operation i.e. drawing off a portion of the concentrated water and replacing it with fresh water.

Scales:

It is hard, sticky substance which deposit firmly on the inner walls of the boiler.scales form non conducting layers. Scales are the main sources of boiler troubles. It can be represented as below.



Causes Of scale formation

1. Decomposition of Calcium Bicarbonate.

$$Ca(HCO_3)_2 \longrightarrow \underset{\text{Scale}}{CaCO_3} + H_2O + CO_2 \downarrow$$

However scale composed chiefly of calcium carbonate is soft and is the main cause of scale formation in **low pressure boiler**. But in high pressure boiler, calcium carbonate is soluble to formation of $Ca(OH)_2$.

$$CaCO_3 + H_2O \longrightarrow Ca(OH)_2 + CO_2 \downarrow$$

2. Decomposition of Calcium Sulphate.

The solubility of calcium sulphate in water decreases with rise in temperature. In other words, $CaSO_4$ is soluble in cold water but almost completely insoluble is super heated water. Consequently, $CaSO_4$ gets precipitated as hard scale on the hot portion of the boiler. This is the main cause of scale in high pressure boiler.

3. Hydrolysis of magnesium salts.

Dissolved magnesium salts gets hydrolyzed (at Prevailing high temperature inside the boiler) forming magnesium hydroxide precipitate, which forms a soft type of scale

$$MgCl_2 + H_2O \rightarrow Mg(OH)_2 \downarrow + 2HCl$$

4. Presence of Silica. Even if a small quantity of SiO_2 is present, it may deposit as calcium

silicate ($CaSiO_3$) and magnesium silicates ($MgSiO_3$).These deposits adhere very firmly on inner side of the boiler surfaces and are very difficult to remove. One important source of silica in water is the sand filter.

Disadvantages / Drawback of Scales formation:

1. Wastage of fuel:

Scales have low thermal conductivity, so the rate of heat transfer from boiler to water decreases therefore more fuel is needed for heating the water.

2. Lowering of boiler safety:

Due to scales, boiler gets overheated to maintain the steady supply of heat. Due to over-heating boiler material becomes softer and weaker and this causes distortion of boiler tubes.

3. Excessive thinning of metal:

Due to overheating, metal becomes weak due to formation of iron oxide thereby causing excessive thinning of metal.

$$3Fe + 4\,H_2O \longrightarrow Fe_3O_4 + 4H_2$$

4. Decreases efficiency of boiler

Scales may sometimes deposit in the valve and condenser of the boiler and chock them partially. This results in decrease in the efficiency of the boiler.

5. Danger of explosion:

Due to cracking of scales and overheating, water comes suddenly in contact with over heated plates. This causes in formation of large amount of steam suddenly.

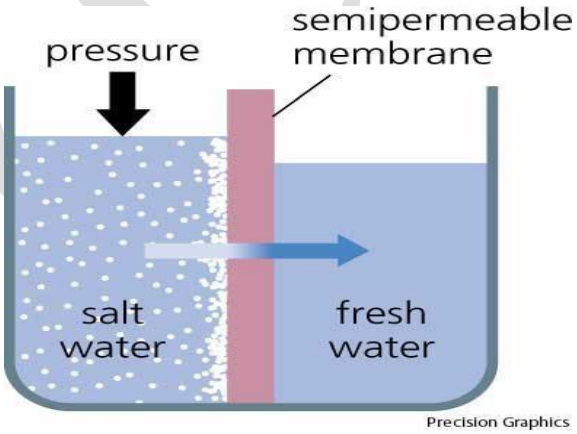
So sudden high pressure is developed which may even cause explosion.

Removal of Scales.

1. With the help of piece of wood or wire brush if they are loosely adhering.
2. Thick scales can be removed by using hammer & chisel
3. By **thermal shocks treatment** i.e. heating the boiler and then suddenly cooling with cold water, if they are brittle.
4. By adding **sodium aluminate** which can trap the scale forming particles.
5. Adding organic chemicals like **tannin** which forms the coating on the scale forming particles.
6. By frequent **blow –down** operation, if they are loosely held.

REVERSE OSMOSIS

1) When two solutions of unequal concentrations are separated by semi permeable membrane, flow of solvent takes place from dilute to concentrated sides, due to osmosis.
2) If a hydrostatic pressure in excess of osmotic pressure is applied on the concentrated side, the solvent flow reverses i.e. solvent is forced to move from concentrated sides to dilute sides across the membrane. This is the principle of reverse osmosis.
3) Thus, in this method pure solvent (water) is separated from its contaminates.
4) In this method, (15-40 kg/cm^2) pressure is applied to the sea/ impure water to force its pure water through the semi-permeable membrane, leaving behind the dissolved solids.
5) The membrane consists of a very thin foil of cellulose acetate.
6) Superior membranes like Polymethacrylate and polyamide polymers are recently used



ADVANTAGES

- It is the best method for water softening.
- The semipermeable membrane will block all ion particles.
- The maintenance of the system is very simple.
- It gives us clean and pure water by blocking all contaminants.
- The available RO systems are very compact, and it requires little space.
- The useful life of the full system, including the membrane, is over two years.
- This system does not require any use of chemicals to purify water.
- The energy requirement for the RO system is very low.
- RO systems are totally automated and are designed to start and stop on their own.

Disadvantages of Reverse Osmosis

Some of the disadvantages of the reverse osmosis process are as follows.

- Sometimes reverse osmosis leads to clogging of the whole system.
- It requires routine filter changes and maintenance.
- The installation cost of a reverse osmosis system is high.
- The whole process is very slow when it comes to household application, as the pressure used is very low.

- The process does not help in disinfecting the water. You will require a separate process to disinfect the water.
- Hard water can damage the system.
- The damaged membrane will allow any small microorganism to pass through it.
- The applied pressure has to be more than the osmotic pressure, or the system won't work.
- The RO system is not self-sustaining.

Applications of Reverse Osmosis

- Reverse osmosis is widely used in residential and commercial water filtration systems. Other than that, it has plenty of applications in various industries. Some of them are mentioned below.
- Reverse osmosis is a type of process that is used to remove dissolved chemical particles from water.
- Reverse osmosis is a type of process that is used to remove dissolved biological entities from water.
- **It is highly used in desalinating seawater.**
- It has crucial applications in the medical field.
- It is used to purify water to prevent any diseases.
- It has a wide application in water treatment and water purification.
- It is used in food industries, and it is applied for the concentration of juices, milk, and other beverages.
- It is used to provide clean water for the community water supply.